Typical Solution for C&I Distributed PV Power Plant

String

ENGLISH

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With the urging need of new energy resources, distributed solar power plant is one of the most popular green energy solution for commercial & industrial application scenarios all around the world. This is typically to build a photovoltaic power plant on top of the building roof or in the open & less-shadowy field to form a "self-use" power plant and even gain more profits from "selling surplus feed-in electricity" to the grid. What's more, the average ROI time period of 1MW solar plant for example, is around 3 years with local subsidies or roughly 5 years without subsidies. And of course it depends on many factors like your local regulation, actual electricity usage and feed-in tariff policy etc.

In this article, we will introduce the latest parallel solution with SolaX Power's C&I string inverters and DataHub1000.

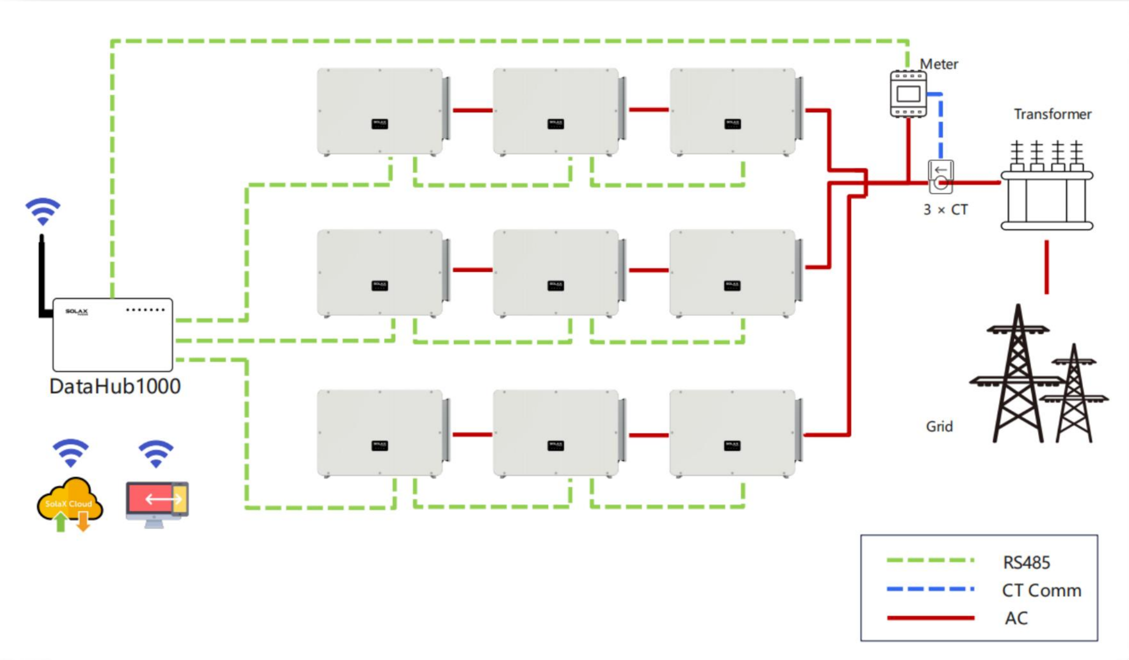
Scenario 1: RS485 Communication Diagram

With all inverters connected to the 3 available RS485 ports on DataHub1000, all the inverter data can be monitored on SolaX Cloud platform. In the near future, based on the Modbus TCP protocol of DataHub1000, it will also be possible to fetch all operating data and achieve remote control on customized platform.

Note: As DataHub1000 has built-in Wi-Fi and LAN, no external monitoring device is required.

For remote control, on SolaX Cloud platform, site manager can easily do operation and maintenance to all inverters 24H x 7.

Furthermore, SolaX provides the inverter Modbus RTU protocol, which is open and accessible to third-party EMS like Solar-log, to further extend more devices control and advanced applications.



Scenario 2: Power Line Communication Diagram

Comparing to the RS485 communication scenario, PLC Box here can save the investment costs on cables and time spent on installations.

For remote control, besides SolaX Cloud, another way is via Ripple Control Receiver(RCR), a receiving device provided by grid operators. RCR usually has 4 digital outputs which means it can support up to 16 different status of DO ports after receiving signals from grid distributors.

DO of RCR will be connected to DI on our DataHub1000, so each status will trigger a scenario of active power control or reactive power control. The 16 status and its action can be set on our DataHub1000 setting page. For more details, please refer to the RCR solution on this link here.

Note: In this scenario, the inverter models be with PLC function

System features

- Support 150% PV oversize to ensure more production even in winter
- String level power management and optimization

- Smart management and maintenance remotely
- Demand Response, Ripple Control and Power Control for grid distributors
- Open API & Modbus & MBUS(PLC) and support third-party platform
- Safety care with dedicated protections

Device List

String Inverters	Datalogger	PLC(Optional)	Meter	Wires
Three-phase 20kW - 150kW	DataHub 1000	PLC Box	DTSU666-CT	PV Cables AC Cables
			CT	Communication Cables

Note: DataHub1000 has built-in Wi-Fi and LAN functions, which is ready to use out of box. Thus no additional Pocket Wi-Fi/LAN is required in this system.

1、String Inverter

X3-MGA-G2 MODEL LIST
X3-MGA-40K-G2
X3-MGA-50K-G2
X3-MGA-60K-G2
X3-MGA-20K-G2-LV
X3-MGA-25K-G2-LV
X3-MGA-30K-G2-LV
X3-MGA-35K-G2-LV

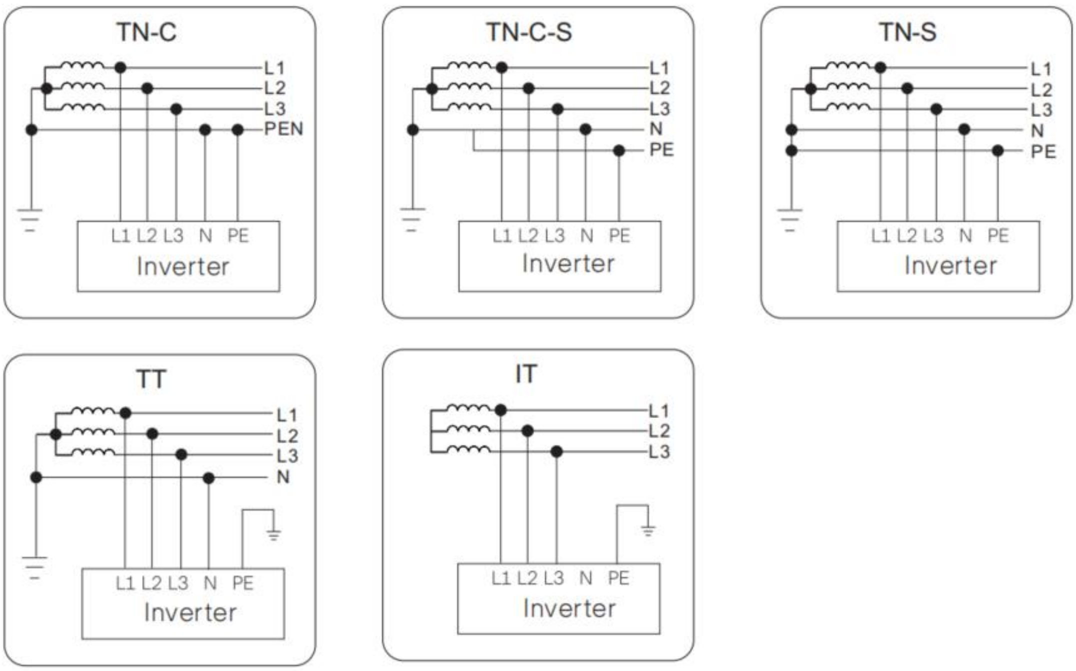
"X3" means Three-phase
"40K" means the nominal AC output is 40kW
"LV" means Low Voltage Range(applicable to 220V Delta Grid)
Without "LV" models are applicable to standard EU&AU Grid type, 3L/~/PE, 3*230V/480Vac.

X3-FTH MODEL LIST
X3-FTH-80K
X3-FTH-100K
X3-FTH-110K
X3-FTH-120K
X3-FTH-125K
X3-FTH-136K-MV
X3-FTH-150K-MV
X3-FTH-40K-LV
X3-FTH-50K-LV
X3-FTH-60K-LV
X3-FTH-70K-LV

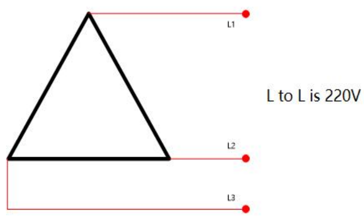
"X3" means Three-phase
"120K" means the nominal AC output is 120kW
"LV" means Low Voltage Range(applicable to 127/220V Delta Grid)
"MV" means Medium Voltage Range(the rated AC output is 540Vac)
Without "LV" or "MV" models are applicable to standard EU&AU Grid type, 3L/~/PE, 3*230V/480Vac.

In accordance with your distributed power plant, one of the key factors to choose the suitable model range is the grid voltage and type. Visit the grid type KB page for more details on this link here.

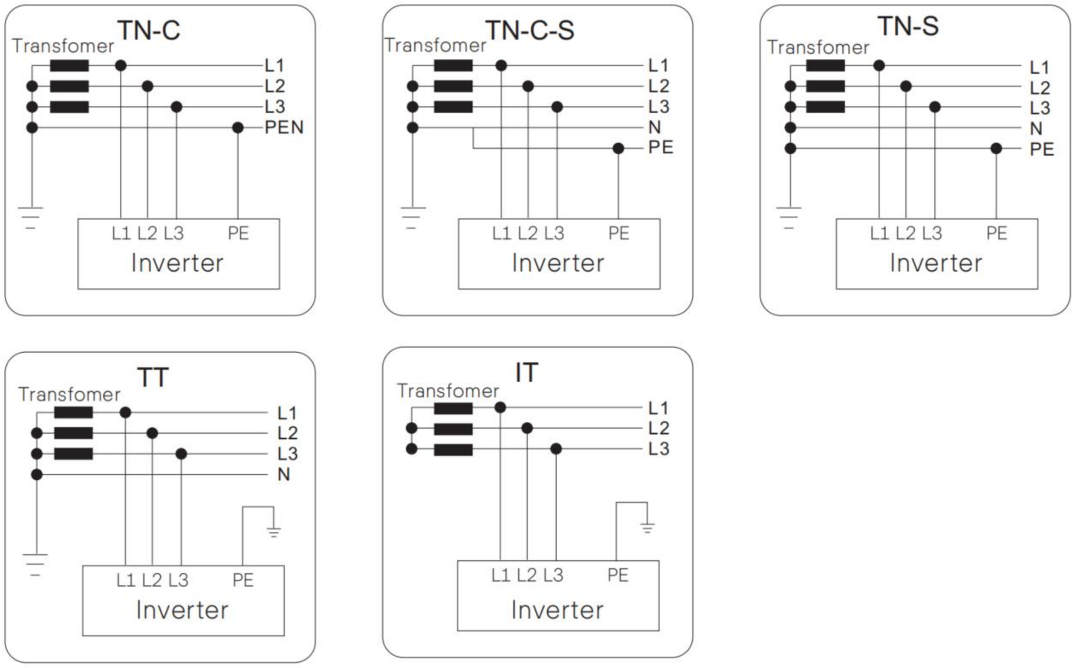
- If the grid is L-N: 220V/230V, L-L: 380V/400V Wye or Delta, the optional models are
  X3-MGA-40K-G2/X3-MGA-50K-G2/X3-MGA-60K-G2;
  X3-FTH-80K/X3-FTH-100K/X3-FTH-110K/X3-FTH-120K/X3-FTH-125K;



- If the grid is 3P/PE 220V Delta, the optional models are
  - X3-MGA-20K-G2-LV/X3-MGA-25K-G2-LV;
  - X3-MGA-30K-G2-LV/X3-MGA-35K-G2-LV;
  - X3-FTH-40K-LV/X3-FTH-50K-LV;
  - X3-FTH-60K-LV/X3-FTH-70K-LV;



- For those High Voltage Grid over 1kV, there should be AC transformer to transform the AC output to the required HV/UHV grid. The optional models are
  X3-MGA-40K-G2/X3-MGA-50K-G2/X3-MGA-60K-G2;
  X3-FTH-80K/X3-FTH-100K/X3-FTH-110K/X3-FTH-120K/X3-FTH-125K;
  X3-FTH-136K-MV/X3-FTH-150K-MV;

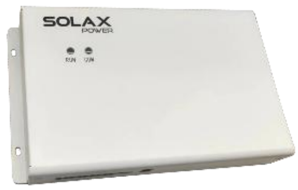


2、DataHub1000

DataHub1000 is a special datalogger for the monitoring platform of photovoltaic power generation system, and it has come together with many functions, with details as follows: interface aggregation, data acquisition, data storage, output control, and centralized monitoring and centralized maintenance of inverters, electricity meters, environmental monitors and other devices in photovoltaic power generation systems.

3、PLC Box

PLC Box is an optional accessory to make power line communication possible for your power plant, with less wiring costs and labor works of connecting communication cables to each inverter and devices. PLC Box can receive the PLC signal via AC power lines, and transmit to DataHub1000 for centralized management.



4、Meter

Meter is an optional accessory for energy measurement and readings. With the smart meter, DTSU666-CT, Datahub1000 can fetch meter readings and transmit data to all inverters connected, also on the monitoring platform the energy data is available.

Note: Meter shall be purchased from SolaX. Meter shall be installed to cover the AC output of all inverters connected to the DataHub1000.

5、CT

CT, also known as current transformer, can measure the current on AC wires, and based on the CT ratio, the readings will be converted to an analog signal for DataHub1000 or inverters to get the actual current values.

Note: The suitable CT size shall be over the maximum possible running current of the AC wires at all times.

We provide the following CT options:

LCTA97C2 200A/5A
LCTA97C2 300A/5A
LCTA97C2 400A/5A

LCTA97C2 600A/5A
ESCT-T36 600A/5A
ESCT-B812 1500A/5A

Note:
CT ratio shall be set correctly on the meter screen, with regards to the CT model.
For instance, the CT ratio shall be 120 if you choose LCTA97C2 600A/5A
Other possible accessories you may use (Need to purchase from other local suppliers)

- **AC Combination Box**

- **Transformer (400V to kV high voltage AC Transformer)**